

Maximal Lotteries: Minimax and Self-Play

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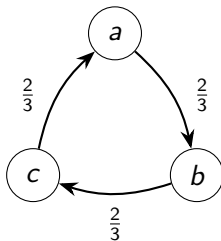
- ▶ In a strict majority cycle, every deterministic output is defeated head-to-head by another completion.
- ▶ Randomization circumvents this: there always exists a lottery that defeats or ties every possible challenger.
- ▶ No-regret learning proves the minimax theorem and gives a way to compute a maximal lottery.
- ▶ Antisymmetry then removes the best-response oracle: one learner can train against its own policy, and its average becomes robust.

Every Single Completion Can Be Beaten

The population contains three equally common preference types:

$$a \succ_1 b \succ_1 c, \quad b \succ_2 c \succ_2 a, \quad c \succ_3 a \succ_3 b.$$

Consequently, $P_{ab} = P_{bc} = P_{ca} = \frac{2}{3}$:

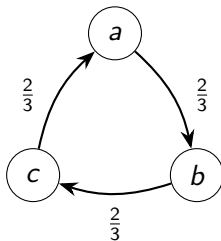


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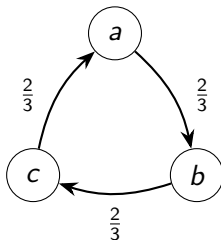
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The obstacle is not the selection rule: this profile contains no single completion that is robust to head-to-head challenge.

Randomization Can Balance the Cycle

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$$p = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right).$$

Against pure challenger a , its probability of winning is

$$\frac{1}{3} (P_{aa} + P_{ba} + P_{ca}) = \frac{1}{3} \left(\frac{1}{2} + \frac{1}{3} + \frac{2}{3} \right) = \frac{1}{2}.$$

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Does every finite comparison problem admit some robust lottery?

Pairwise Margins Put a Tie at Zero

Score a comparison as $+1$ when i wins and -1 when j wins. Its expected score is

$$M_{ij} := (+1)P_{ij} + (-1)P_{ji} = P_{ij} - P_{ji} = 2P_{ij} - 1.$$

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Reversing the two completions reverses the margin:

$$M_{ji} = -M_{ij}, \quad M = -M^{\top}.$$

In particular, every completion has zero margin against itself:
 $M_{ii} = 0$.

General Policies Can Be Compared Pairwise

Let $p, q \in \Delta(C)$ be policies. Draw $i \sim p$ and $j \sim q$. Their expected win-minus-loss margin is

$$\mathbb{E}_{i \sim p, j \sim q}[M_{ij}] = \sum_{i, j \in C} p_i q_j M_{ij} = p^\top M q.$$

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Since $P_{ij} = (1 + M_{ij})/2$, the probability that p wins is

$$\sum_{i, j \in C} p_i q_j P_{ij} = \frac{1}{2} + \frac{1}{2} p^\top M q.$$

Thus p defeats or ties q exactly when $p^\top M q \geq 0$.

A Robust Policy is Evaluated Against the Strongest Challenger

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If Max commits to p , Min chooses the most damaging challenger q :

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Max therefore seeks the policy with the best worst-case margin:

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The opponent distribution is not fixed in advance; it responds strategically to the proposed policy.

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Max can never guarantee a strictly positive margin.

But does some policy guarantee margin zero against *every* challenger?

The Robustness Question Is a Zero-Sum Game

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For the preference game, $G = M$ and $d_{\max} = d_{\min} = |C|$, so $\Delta_{d_{\max}}$ and $\Delta_{d_{\min}}$ are both $\Delta(C)$. We develop the theory for general G and then return to M .

Commitment Order Defines Two Values

If Max commits first, Max guarantees the *maximin value*

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The responding player sees the commitment before choosing a best response. Should committing first be costly?

Committing First Cannot Be an Advantage

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Thus

$$v_{\text{maximin}} \leq v_{\text{minimax}}.$$

This is *weak minimax*. The question is whether the gap can be positive.

With Randomization, Commitment Order Does Not Matter

Theorem (von Neumann's minimax theorem)

For every finite payoff matrix $G \in \mathbb{R}^{d_{\max} \times d_{\min}}$,

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We will prove this by imagining repeated play: Min learns from experience, while Max best-responds to Min's current strategy.

Proof Roadmap

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2. Let Min run Hedge while Max best-responds to Min's current strategy.
3. Bound their average payoff below by v_{minimax} and above by $v_{\text{maximin}} + \text{Reg}_T / T$. Vanishing regret closes the commitment gap.

The same repeated play will also *compute* the robust lottery.

An Online Learner Faces an Adversary

In rounds $t = 1, \dots, T$:

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The adversary is unrestricted: the loss sequence may depend on everything the learner has done. What guarantee could possibly survive that?

External Regret Compares Against the Best Fixed Action

The learner's *external regret* is

$$\text{Reg}_T := \sum_{t=1}^T z_t^\top \ell_t - \min_{z \in \Delta_d} \sum_{t=1}^T z^\top \ell_t.$$

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The Hedge Algorithm

Fix a learning rate $\eta > 0$ and initialize $h_{1,a} = 1$ for every action a .
In each round t :

1. Play the distribution

$$z_{t,a} := \frac{h_{t,a}}{\sum_b h_{t,b}}.$$

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Lower-loss actions retain more relative weight; η controls how aggressively the distribution responds.

Hedge Competes with Every Fixed Action

Theorem (Hedge regret bound)

For every sequence $\ell_1, \dots, \ell_T \in [-1, 1]^d$ and every $\eta > 0$, Hedge satisfies

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Choosing η of order $1/\sqrt{T}$ makes Reg_T / T vanish. We now prove the bound.

The Analysis Tracks Cumulative Weight

Let

$$Z_t := \sum_a h_{t,a}, \quad \hat{L}_T := \sum_{t=1}^T z_t^\top \ell_t, \quad L_{T,a} := \sum_{t=1}^T \ell_{t,a}.$$

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Comparing the bounds will compare the learner with every fixed action.

The Weight Ratio Is an Exponential Moment

The Hedge update gives

$$\begin{aligned} Z_{t+1} &= \sum_a h_{t,a} e^{-\eta \ell_{t,a}} \\ &= Z_t \sum_a z_{t,a} e^{-\eta \ell_{t,a}}. \end{aligned}$$

Hence

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If $A \sim z_t$ and $Y = -\ell_{t,A}$, then

$$\mathbb{E}[e^{\eta Y}] = \sum_a z_{t,a} e^{-\eta \ell_{t,a}} = \frac{Z_{t+1}}{Z_t}, \quad \mathbb{E}[Y] = -z_t^\top \ell_t.$$

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So the one-round growth of total weight is exactly an exponential moment of a bounded random variable. We need a tool that bounds such moments.

Hoeffding's Lemma Bounds the Exponential Moment

Lemma (Hoeffding's lemma)

If $Y \in [-1, 1]$, then for every $\eta > 0$,

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This is exactly the form we need: on the previous slide, $Y = -\ell_{t,A}$ takes values in $[-1, 1]$ and its exponential moment is the weight ratio Z_{t+1}/Z_t .

Learner Loss Upper-Bounds Final Total Weight

Substituting the two identities into Hoeffding's lemma gives

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Summing over t telescopes the logarithms:

$$\begin{aligned} \log Z_{T+1} - \log Z_1 &\leq -\eta \sum_{t=1}^T \mathbf{z}_t^\top \ell_t + \frac{\eta^2 T}{2} \\ &= -\eta \hat{L}_T + \frac{\eta^2 T}{2}. \end{aligned}$$

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Since $Z_1 = d$,

$$\log Z_{T+1} \leq \log d - \eta \hat{L}_T + \frac{\eta^2 T}{2}.$$

Any Fixed Action Lower-Bounds Final Total Weight

Fix an action a . Iterating its Hedge update gives

$$\begin{aligned} h_{T+1,a} &= \prod_{t=1}^T e^{-\eta \ell_{t,a}} \\ &= e^{-\eta L_{T,a}}. \end{aligned}$$

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$$\begin{aligned} h_{T+1,a} &= \prod_{t=1}^T e^{-\eta \ell_{t,a}} \\ &= e^{-\eta L_{T,a}}. \end{aligned}$$

Since Z_{T+1} is the sum of all final weights,

$$Z_{T+1} \geq h_{T+1,a}.$$

Taking logarithms therefore gives

$$\log Z_{T+1} \geq -\eta L_{T,a}.$$

The Two Weight Bounds Give Low Regret

Combining the upper and lower bounds gives

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This holds for every action a , so

$$\boxed{\text{Reg}_T \leq \frac{\log d}{\eta} + \frac{\eta T}{2}.$$

Balancing the Two Terms Gives Vanishing Average Regret

Choose

$$\eta = \sqrt{\frac{2 \log d}{T}}.$$

Then

$$\frac{\log d}{\eta} = \frac{\eta T}{2} = \sqrt{\frac{T \log d}{2}}.$$

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Therefore

$$\text{Reg}_T \leq \sqrt{2 T \log d}, \quad \frac{\text{Reg}_T}{T} \leq \sqrt{\frac{2 \log d}{T}} \rightarrow 0.$$

Hedge is a no-regret algorithm.

A Thought Experiment: Min Learns, Max Best-Responds

Repeat the game for T rounds:

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Min's expected loss is $q_t^\top \ell_t = p_t^\top G q_t$.

Write $u_t := p_t^\top G q_t$ for this payoff to Max. We will bound the same average payoff from below and above.

Best Responses Keep Payoff Above the Minimax Value

Because p_t is a best response to q_t ,

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Consequently,

$$\frac{1}{T} \sum_{t=1}^T u_t \geq v_{\text{minimax}}.$$

Min's Regret Keeps Payoff Near the Maximin Value

Let $\text{Reg}_T^{\text{Min}}$ denote Min's external regret on the loss vectors $\ell_t = G^\top p_t$. The regret guarantee says

$$\sum_{t=1}^T u_t - \min_q \sum_{t=1}^T p_t^\top Gq \leq \text{Reg}_T^{\text{Min}}.$$

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The first term is the worst-case payoff of the specific policy $\bar{p}_T$, and v_{maximin} is by definition the largest such worst-case payoff over all policies:

$$\frac{1}{T} \sum_{t=1}^T u_t \leq v_{\text{maximin}} + \frac{\text{Reg}_T^{\text{Min}}}{T}.$$

Vanishing Regret Closes the Commitment Gap

The lower and upper bounds on the average payoff give

$$v_{\text{minimax}} \leq \frac{1}{T} \sum_{t=1}^T u_t \leq v_{\text{maximin}} + \frac{\text{Reg}_T^{\text{Min}}}{T}.$$

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Hedge gives $\text{Reg}_T^{\text{Min}} / T \rightarrow 0$. Letting $T \rightarrow \infty$ proves

$$v_{\text{maximin}} = v_{\text{minimax}}.$$

Antisymmetry Establishes the Value of the Preference Game

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Minimax says these two values are equal. Therefore both equal zero:

$$\boxed{\max_p \min_q p^T M q = 0.}$$

A Maximal Lottery Always Exists

Because the value is zero, some $p^* \in \Delta(C)$ satisfies

$$\min_q (p^*)^\top Mq = 0.$$

Equivalently,

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Equivalently,

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Such a policy is called a *maximal lottery*. Its probability of winning against any challenger is at least

$$\frac{1}{2} + \frac{1}{2}(p^*)^\top Mq \geq \frac{1}{2}.$$

The Minimax Proof Uses a Best-Response Oracle

The proof suggests an algorithm: Min learns with Hedge, while Max returns an exact best response p_t on every round. Averaging those responses gives

$$\min_q \bar{p}_T^\top M q \geq -\frac{\text{Reg}_T^{\text{Min}}}{T}, \quad \bar{p}_T := \frac{1}{T} \sum_{t=1}^T p_t.$$

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Can antisymmetry eliminate the best-response oracle?

Self-Play Eliminates the Best-Response Oracle

On round t , one learner chooses $p_t \in \Delta(C)$. Instead of constructing a separate best response, evaluate every action against opponents drawn from p_t itself by giving the learner the loss vector

$$\ell_t := -Mp_t.$$

Thus action a has loss $\ell_{t,a} = -(Mp_t)_a$: actions that tend to defeat the current policy have lower loss.

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Antisymmetry gives $p_t^\top Mp_t = 0$, and hence

$$p_t^\top \ell_t = 0 \quad \text{on every round.}$$

Regret will measure whether some fixed challenger could have systematically beaten the learner's changing policies.

External Regret Bounds Every Fixed Challenger

$$\underbrace{\sum_{t=1}^T p_t^\top \ell_t}_{\text{learner's cumulative loss}} - \underbrace{\min_{q \in \Delta(C)} \sum_{t=1}^T q^\top \ell_t}_{\text{best fixed challenger's cumulative loss}} \leq \text{Reg}_T.$$

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The self-tie identity makes the learner's cumulative loss zero. Substituting $\ell_t = -Mp_t$ therefore gives

$$\text{Reg}_T \geq -\min_q \sum_{t=1}^T q^\top (-Mp_t) = \max_q \sum_{t=1}^T q^\top Mp_t,$$

where the last equality uses $-\min_q f(q) = \max_q [-f(q)]$.

A Fixed Challenger Sees Only the Average Policy

For any fixed challenger q , linearity gives

$$\sum_{t=1}^T q^\top M p_t = q^\top M \left(\sum_{t=1}^T p_t \right) = T q^\top M \bar{p}_T, \quad \bar{p}_T := \frac{1}{T} \sum_{t=1}^T p_t.$$

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Taking the maximum over q in the preceding regret bound gives

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Define the *exploitability* of p by

$$\text{Expl}(p) := \max_{q \in \Delta(C)} q^\top M p.$$

A policy is a maximal lottery exactly when $\text{Expl}(p) = 0$.

$$\text{Expl}(\bar{p}_T) \leq \frac{\text{Reg}_T}{T}.$$

Hedge Is an Abstraction of Policy Training

The self-play theorem is exact for a finite, explicitly represented set:

$$p_t \in \Delta(C), \quad \ell_t(y) = -(Mp_t)_y.$$

Tabular Hedge stores one weight and constructs one loss coordinate for every $y \in C$. When C is the set of language-model completions, this is not a scalable algorithm.

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Self-play has nevertheless made an important simplification:

one learner facing itself replaces
a learner plus a best-response oracle

To obtain a scalable procedure, we next replace the tabular learner by a sampled, parameterized policy optimizer.

Sampled Policy Updates Implement the Self-Play Idea

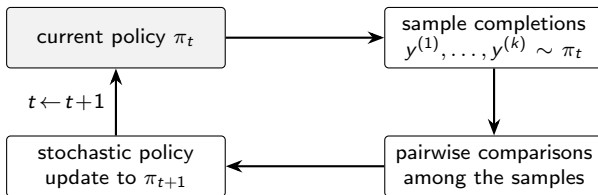
For a prompt x , sample completions from a parameterized policy $\pi_t(\cdot \mid x)$. Pairwise comparisons among the samples estimate the on-policy reward

$$r_t(x, y) := \mathbb{E}_{y' \sim \pi_t(\cdot \mid x)}[M_x(y, y')].$$

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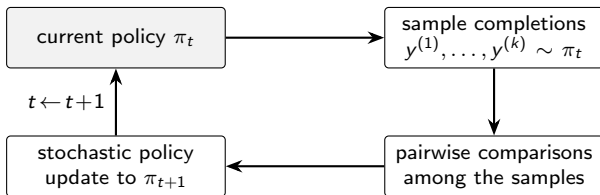
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Sampling and parameter sharing avoid explicit enumeration. The robustness theorem is conditional on the policy optimizer achieving low regret on these changing objectives.

Summary

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- ▶ A strict cycle makes every single completion vulnerable, while a lottery can balance the competing head-to-head defeats.
- ▶ Hedge achieves $O(\sqrt{T \log d})$ external regret by comparing total weight with the weight of each fixed action.
- ▶ If Min learns while Max best-responds, their average payoff is trapped between the minimax and maximin values up to Min's regret.
- ▶ Minimax proves that every finite comparison problem admits a maximal lottery.
- ▶ Antisymmetry permits self-play: one no-regret learner satisfies $\text{Expl}(\bar{p}_T) \leq \text{Reg}_T / T$ without an exact best-response oracle.

Thanks!

See you next class!

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